Introduction to Programming

Lecture No. 8

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In the last lecture

- Loops
  - While
  - Do while
  - For
- Operators
  - Increment / Decrement
  - Compound Assignment Operators

Example: Program to calculate the average marks of class

```
int sum;
int students ;
int average ;
sum = 0 ;
students = 0 ;
do
{
 cin >> grade ;
 sum += grade ;
 students ++ ;
}
while (grade >= 0) ;
average = sum / students ;
cout << average ;
```

A Flaw in the code

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Multi-way decision

if Statements

```
if  ( grade =='A' )
 cout << " Excellent " ;
if  ( grade =='B' )
 cout << " Very Good " ;
if  ( grade =='C' )
 cout << " Good " ;
if  ( grade =='D' )
 cout << " Poor " ;
if  ( grade =='F' )
 cout << " Fail " ;
```

if else

```
if  ( grade ==‘A’ )
 cout << “ Excellent ” ;
else
 if  ( grade ==‘B’ )
  cout << “ Very Good ” ;
else
 if  ( grade ==‘C’ )
  cout << “ Good ” ;
else
 if  ( grade ==‘D’ )
  cout << “ Poor ” ;
```

if else

```
if ( grade == 'A' )
 cout << " Excellent " ;
else if ( grade == 'B' )
 ...
else if ...
 ...
else ...
```

switch statement

switch statements

```
switch ( variable name )
   {
    case 'a' :
     statements;
   case 'b' :
     statements;
 case 'c' :
     statements;

 ...
  }
```

switch statements

```
switch ( grade)
  {
   case 'A' :
    cout << " Excellent " ;
   case 'B' :
    cout << " Very Good " ;
   case 'C' :
    ...
   ...
  }
```

switch statements

```
case 'A' :
  cout << " Excellent " ;
  ...
  ...
```

Example

```
switch ( grade)
{
    case 'A' :
     cout << " Excellent " ;
    case 'B' :
     cout << " Very Good " ;
    case 'C' :
   cout << "Good " ;
   case 'D' :
     cout << " Poor " ;
    case 'F' :
     cout << " Fail " ;
}
```

break;

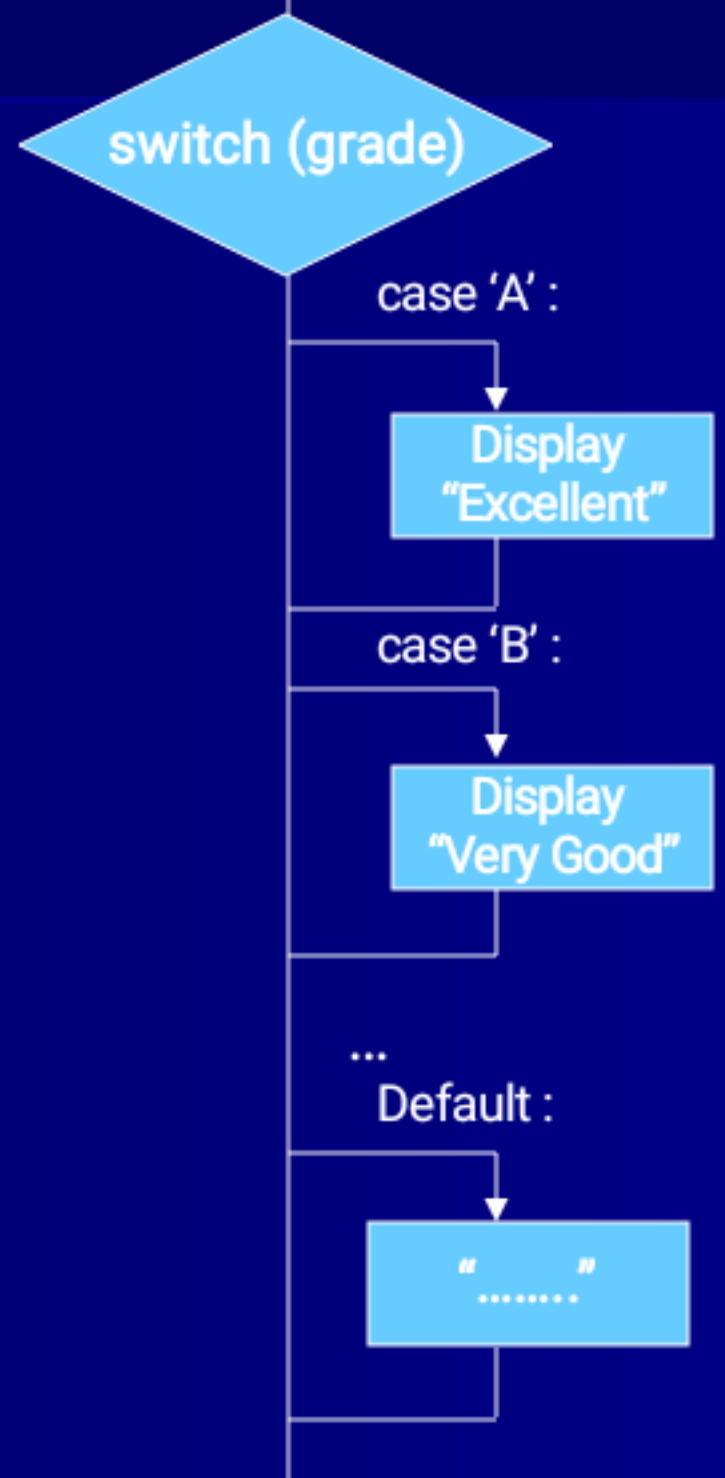
Example

```
switch ( grade )
{
      case 'A' :
   cout << " Excellent " ;
   break ;
      case 'B' :
       cout << " Very Good " ;
    break ;
      case 'C' :
   cout << "Good " ;
    break ;
   case 'D' :
       cout << " Poor " ;
    break ;
      case 'F' :
       cout << " Fail " ;
    break ;
}
```

default :

```
default :
cout << " Please Enter Grade from 'A'
  to 'D' or  'F'  " ;
```

Flow Chart of switch statement



```
if ( amount > 2335.09 )
        statements ;
```

Whole Number

- short
- int
- long

case ‘A’ :

case ‘ 300 ‘ :

case ‘ f ‘ :

break ;

```
if (c == 'z' )

{
 cout << " Great ! You have made the correct guess " ;
     break ;
}
```

continue ;

continue

```
while trynum <= 5 ;
{
    ....
    ....
    continue ;
}
```

continue in ‘for’ loop

```
for ( counter = 0 ;counter <= 10 ; counter ++ )
{
 .......
 continue ;
}
```

What have we done till now …

- Sequential Statements
- Decisions
  - if , if else , switch
- Loops
  - while , do while , for

goto

Unconditional Branch of Execution

Structured Programming

- Sequences
- Decisions
- Loops

- Minimize the use of break
- Minimize the use of continue
- Never use goto

Guide lines for structured programming

- Modular
- Single entry - single exit

Rules for Structured Flowchart

Rule 1 : Use the simplest flowchart

Rule 2 : Any rectangle can be replaced by two rectangles.

Rule 3 : Any rectangle can be replaced with structured flowcharting constructs.

Rule 4 : It says, rule 2 and rule 3 can be repeated as many times as needed

Next Milestones

- Data Structures
  - Arrays
- Character Strings
- Pointers